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mahin-mr12 commit

9eed2b6 · 3 weeks ago



341 lines (341 loc) · 6.41 KB

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## Experiment 7: J Functions

1. Write a Python function to find the maximum and minimum numbers from a sequence of numbers. (Note: Do not use built-in functions.)

```
In [ ]: def find_max_min(numbers):  
        if len(numbers) == 0:  
            return None  
  
        max_num = numbers[0]  
        min_num = numbers[0]  
  
        for num in numbers:  
            if num > max_num:  
                max_num = num  
            if num < min_num:  
                min_num = num  
  
        return max_num, min_num  
  
nums = [10, 6, 8, 90, 12, 56]  
print(find_max_min(nums))
```

(90, 6)

2. Write a Python function that takes a positive integer and returns the sum of the cube of all the positive integers smaller than the specified number.

```
In [ ]: def sum_of_cubes(n):  
        total = 0  
        for i in range(1, n):  
            total += i ** 3  
        return total
```

```
print(sum_of_cubes(5))
```

100

3. Write a Python function to print 1 to n using recursion. (Note: Do not use loop)

In [ ]:

```
def print_numbers(n):  
    if n == 0:  
        return  
    print_numbers(n - 1)  
    print(n)
```

```
print_numbers(5)
```

1  
2  
3  
4  
5

4. Write a recursive function to print Fibonacci series upto n terms.

In [ ]:

```
def fibonacci(n):  
    if n <= 1:  
        return n  
    return fibonacci(n - 1) + fibonacci(n - 2)
```

```
def print_fibonacci(n):  
    for i in range(n):  
        print(fibonacci(i), end=" ")
```

```
print_fibonacci(7)
```

0 1 1 2 3 5 8

5. Write a lambda function to find volume of cone.

```
In [ ]: import math

volume_cone = lambda r, h: (1/3) * math.pi * r**2 * h

print(volume_cone(3, 5))
```

47.12388980384689

6. Write a lambda function which gives tuple of max and min from a list.

Sample input: [10, 6, 8, 90, 12, 56]

Sample output: (90,6)

```
In [ ]: max_min = lambda lst: (
        max(lst), min(lst)
    )

# Sample input
data = [10, 6, 8, 90, 12, 56]
print(max_min(data))
```

(90, 6)

7. Write functions to explain mentioned concepts:

- a. Keyword argument
- b. Default argument
- c. Variable length argument

```
In [ ]: #a)
def student(name, age):
    print("Name:", name)
    print("Age:", age)

student(age=20, name="Mahin")
```

Name: Krish

Age: 20

In [ ]:

```
#b)
def greet(name="Guest"):
    print("Hello", name)

greet("Mahin")
greet()
```

Hello Krish

Hello Guest

In [ ]:

```
#c)
def total_marks(*marks):
    total = 0
    for m in marks:
        total += m
    print("Total:", total)

total_marks(10, 20, 30)
```

Total: 60